

D.I.V.A: Spatial Navigation for the Visually Impaired using Depth Sensing Artificial Intelligence, Convolutional Neural Networks, and Sensor Fusion

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Visual impairment remains a challenging issue in biomedical sciences with a need for universally effective and affordable solutions. Severe conditions like cataracts, glaucoma, macular degeneration, and complete blindness due to visual cortex damage continue to lack economic remedies. Current solutions such as the white cane and optical surgeries are restrictive in mobility, invasive, or costly. To address this, we propose D.I.V.A, a novel wearable device to assist the visually impaired in unfamiliar environments. The device utilizes a stereoscopic camera, RGB camera, gyroscope, custom PCB, electromagnetic-solenoid tactile interface, and proprietary computer vision software. Through a custom U-net neural network trained with 80 object-class types, the device detects and classifies obstacles in the user's surrounding environment. In addition, it communicates this information through a GUI-commanded personalized spatial audio infrastructure, in conjunction with indicative actuation of the tactile interface. The proposed aid is remarkably affordable, costing around \$250 - \$300, making it accessible to individuals facing financial constraints. To justify the proposed system and measure its performance, successful experiments were conducted in indoor environments. The experimental results demonstrate that the proposed assistive device performs all its functions with high accuracy allowing visually impaired people to feel safe and comfortable in an indoor environment. We envision this device significantly improving the lives of visually impaired individuals, especially in regions with limited access to expensive solutions.